SYNTHETIC HOMOTOPY THEORY AND THE SEGAL CONJECTURE

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Abstract. These notes are for the talks at AMSS during 2025 summer. We will recall some basic definition in equivariant homotopy theory. We will also review the synthetic homotopy theory. Then we will give a proof of Segal conjecture for large objects.

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Introduction

There are two processes in equivariant homotopy theory. The first one is "completion in a geometrical sense", that is, to make a space free by production with EG, a model for a free contractable G-space. The second one is "completion in an algebraic sense", that is, to do I-completion for a ring object, usually the 0-th homotopy ring of a ring spectrum, with respect to its augmented ideal. In many cases there will be a comparison map between these two processes, but to determine whether the map is an isomorphism is in general complicated.

The inspiration for Segal conjecture arose from such an example. We consider the equivariant K-theory for a compact Lie group G. The projection $p : EG \to *$ induces a map

$$p^* \colon \mathrm{K}_G^*(X) \to \mathrm{K}_G^*(EG_+ \wedge X)$$

Note that the G-equivariant K-theory is always an $RO(G) \cong \mathrm{K}_G^*(*)$-module. We denote by $I = \ker(RO(G) \to \mathbb{Z})$ the augmented ideal. We have the following Atiyah-Segal completion theorem.

Theorem 0.1 (Atiyah-Segal)**.** *For X a finite G-CW complex, the map p^* is the completion at the augmentation ideal I. Thus we have*

$$\mathrm{K}_G^*(X)^\wedge_I \cong \mathrm{K}_G^*(EG_+ \wedge X)$$

The Segal conjecture arose from such a belief that there should by some analogy between $\mathrm{K}_G^*(-)$ and $\pi_G^*(-)$. We want to replace $\mathrm{K}_G^*(-)$ by $\pi_G^*(-)$. In order to do this, we first need to replace $RO(G)$ by $\pi_G^0(\mathbb{S})$ and I by its counterpart in stable cohomology theory.

It is an insight from Segal that there is an isomorphism between $\pi_G^0(\mathbb{S})$ and $A(G)$, which we will refer to as the *Burnside ring*. One can similarly define an augmented map $A(G) \to \mathbb{Z}$ and take I to be the kernel. As a result, one can copy every word in Atiyah-Segal completion theorem and conjecture that there should be a similar equivalence for G-stable homotopy theories. The conjecture was brought up by Segal, motivating a series of works [Lin80], [Gun80], [MM82] and was eventually solved by Carlsson [Car84].

Theorem 0.2 (Segal-Carlsson)**.** *For any finite group G, there is an equivalence of G-spectra*

$$\xi^* \colon \mathbb{S}_I^\wedge \simeq \mathrm{map}(EG_+, \mathbb{S})$$

In particular, for any finite G-spectrum X, we have an isomorphism

$$\pi_G^*(X)_I^\wedge \cong \pi_G^*(EG_+ \wedge X)$$

and take $X = \mathbb{S}$ we have $A(G)_I^\wedge \cong \pi_G^0(EG_+)$.

We can also generalize this result to family of subgroups. Let $\mathscr{O}_G$ denote the category of G-orbits. By Elmendorf's theorem, we have an equivalence

$$\Psi \colon \mathrm{Fun}(\mathscr{O}_G, \mathcal{S}) \simeq G\mathcal{S} \colon \Phi$$

where Φ sends X to $\Phi(X)$, which takes G/H to X^H. Recall we have the following definition

Definition 0.3. Let G be a compact Lie group.

(1) A *family* $\mathscr{F}$ in G is a set of subgroups of G such that if $H \in \mathscr{F}$ and some conjugacy of K is contained in H, then $K \in \mathscr{F}$.
(2) For a family $\mathscr{F}$, $\underline{\mathscr{F}} \in \mathrm{Fun}(\mathscr{O}_G, \mathcal{S})$ is defined by

$$\underline{\mathscr{F}}(G/H) = \begin{cases} * & \text{for } H \in \mathscr{F} \\ \emptyset & \text{otherwise} \end{cases}$$

We define the *universal $\mathscr{F}$-space* as $E\mathscr{F} := \Psi(\underline{\mathscr{F}})$.

It follows immediately from the definition that for $\mathscr{F} = \{e\}$, $E\mathscr{F} = EG$ is the classical universal space. The following generalization is given by May.

Theorem 0.4 (May)**.** *For any finite group G and any family $\mathscr{F}$, there is an equivalence of G-spectra*

$$\xi^* \colon \mathbb{S}_{I\mathscr{F}}^\wedge \simeq \mathrm{map}(E\mathscr{F}_+, \mathbb{S})$$

A well known corollary of the Segal conjecture, which is sometimes also referred to as the Segal conjecture, states that the canonical map $\mathbb{S} \to \mathbb{S}^{tC_p}$ inherits $\mathbb{S}^{tC_p}$ as the p-completion of $\mathbb{S}$. The same is ture if one replace $\mathbb{S}$ by $X \in \mathrm{Sp}$ in the thick subcategory generated by $\mathbb{S}$ since the Tate construction is exact. Recently using some input from synthetic homotopy theory, [BS24] proved the Segal conjecture for larger objects:

Theorem 0.5 (Theorem 3.3 ; [BS24])**.** *Let I be the augmented ideal of the mod p Steenrod algebra. Let X be a p-complete, bounded below spectrum with I-complete cohomology, then the canonical map*

$$X \to X^{tC_p}$$

is an equivalence.

The aim of these talks is to explain the proof of the Segal conjecture and its corollaries. These notes are divided as follows. In Section 1, we will outline the classical proof of the Segal conjecture using (non-equivariant) Adams spectral sequence. We begin with some recollections of basic concepts and explain the proof for general families. Then we explain its corollary and relationship with the Tate construction. In Section 2, we will recall some basic inputs from synthetic homotopy theory, setting the stage for understanding the modern proof of [BS24]. In Section 3 we will explain the proof of [BS24].

1. The classical story

In this part, we recall some basic definitions and facts that will be needed in the outline of Segal conjecture. We will give a sketch of Carlsson's classical proof using computation of Adams spectral sequence. And at the end of this section, we will recall the Tate construction for finite p-groups.

1.1. Burnside ring.

We recollect some key results of the Burnside ring. The main reference for this part is ??? and most proof will be skipped.

Let G be a compact Lie group. In this section, we will quickly give an insight into the basic properties of the Burnside ring $A(G)$. Before define this ring, we will need the following description of (strong) dualizable G-spectrum

Definition 1.1. Let $\mathscr{C}$ be a symmetric monoidal category, a dualizable object in $\mathscr{C}$ is an object X together with another object $X^\vee$, called the dual of X, such that there are maps $\epsilon : X^\vee \wedge X \to \mathcal{I}$ and $\eta : \mathcal{I} \to X \wedge X^\vee$, called evaluation and coevaluation maps, making the following diagrams commute

$$\begin{array}{ccc} X & \xrightarrow{\eta\wedge\mathrm{id}} & X \wedge X^\vee \wedge X \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \mathrm{id}\wedge\epsilon} \\ & & X \end{array} \qquad \begin{array}{ccc} X^\vee & \xrightarrow{\mathrm{id}\wedge\eta} & X^\vee \wedge X \wedge X^\vee \\ & \searrow^{\mathrm{id}} & \downarrow{\scriptstyle \epsilon\wedge\mathrm{id}} \\ & & X \end{array}$$

We will focus on the category $G\mathrm{Sp}$. That is, the G-equivariant stable category. We let $\mathrm{map(X, Y)}$ denote the mapping spectrum of two objects.

Definition 1.2. Let E be a G-spectrum. We say that E is strongly dualizable if the natural map $\nu : E \wedge \mathrm{map}(E, \mathbb{S}) \to \mathrm{map}(E, E)$ is an isomrophism.

One can show that a G-spectrum E is dualizable in the sense of Definition 1.1 if and only if it is strongly dualizable in the sense of Definition 1.2. In this case, we have $E^\vee \cong \mathrm{map}(E, \mathbb{S})$. In fact, this is a vary standard argument, see for example, [HM24, Lemma B.1.16].

All the spectra are not dualizable. However, it is absolutely clear what the dualizable spectra would be like.

Lemma 1.3. *A G-CW spectrum is dualizable if and only if it is a summand of a finite G-CW spectrum.*

It is natural to define the Euler characteristic of a dualizable object in a symmetric monoidal category.

Definition 1.4. Let E be a strongly dualizable G-spectrum. The Euler characteristic of E is defined to be the composite

$$\chi(E) : \mathbb{S} \xrightarrow{\eta} E \wedge E^{\vee} \xrightarrow{\cong} E^{\vee} \wedge E \xrightarrow{\epsilon} \mathbb{S}$$

We list some properties of the Euler characteristic

Proposition 1.5. *The Euler characteristic satisfies the following properties.*
(a) $\chi(*)$ *is the constant map,* $\chi(\mathbb{S})$ *is the identity map;*
(b) $\chi(X) = \chi(Y)$ *if* X *is* G*-equivalent to* Y*;*
(c) $\chi(X \vee Y) = \chi(X) + \chi(Y)$, $\chi(X \wedge X) = \chi(X)\chi(Y)$*;*
(d) $\chi(\Sigma^n X) = (-1)^n \chi(X)$ *and for* $f : X \to Y \to Cf$ *a cofiber sequence, we have* $\chi(Cf) = \chi(Y) - \chi(X)$*;*
(e) For E *a finite* G*-CW spectrum, we have*

$$\chi(X) = \sum_n \sum_{(H)} (-1)^n \nu(H, n) \chi(G/H)$$

where $\nu(H, n)$ *denote the number of* n*-cells of type* G/H*.*

Remark 1.6. (Why Euler classes are called Euler classes)

We now give a generalized definition of the Burnside ring. This definition is due to Tom Dieck.

Definition 1.7 (Tom Dieck)**.** Let G be a compact Lie group. The Burnside ring $A(G)$ is defined to be the set of the equivalent classes of compact G-Euclidean neighborhood retracts. The equivalence is given by $X \sim Y$ if and only if $\chi(X) = \chi(Y)$. The sum and product are induced by the disjoint union and wedge product.

A result in non-stable equivariant homotopy theory states that, two self-maps f_1, f_2 of a G-representative sphere are homotopy if and only if for every H such that WH is finite, $\deg(f_1^H) = \deg(f_2^H)$. We can therefore conclude that the Burnside ring $A(G)$ is a free Abelian group with generators $[G/H]$ for every conjugacy class (H) with WH finite. When G is finite, this is equivalent to the Grothendick ring associated to the set of isomorphism classes of finite G-sets.

The structure of $A(G)$ is rather complicated for a general compact Lie group. In order to describe it, we will construct another ring $C(G)$. Let $\Xi(G)$ denote the set of conjugacy classes of subgroups H of G such that WH is finite. $\Xi(G)$ is equipped with a topology, which is induced from the usual distance topology of subgroups.

The first thing to say, is that $\Xi(G)$ somehow controls the finiteness of $A(G)$.

Proposition 1.8. *Let* G *be a compact Lie group. Let* T *be the maximal torus of* G*. Then the followings are equivalent.*
(1) $\Xi(G)$ *is finite.*
(2) WT *acts trivially on* T*.*
(3) $A(G)$ *is Noetherian.*

Definition 1.9. Let $\Xi(G)$ be as above.

(1) We define $C(G)$ as the space of continuous functions $\Xi(G) \to \mathbb{Z}$. Note that since $\Xi(G)$ is finite, these functions always take finitely many values.
(2) We define $\chi \colon A(G) \to \pi_0^G(\mathbb{S})$ as the Euler characteristic of the suspension spectrum as defined in Definition 1.4.

(3) We define $d\colon \pi_0^G(\mathbb{S}) \to C(G)$ by

$$d(f)(H) = \deg(f^H)$$

Therefore, we have a commutative diagram

$$\begin{array}{ccc} A(G) & \xrightarrow{\quad\chi\quad} & \pi_0^G(\mathbb{S}) \\ & {\scriptstyle\phi}\searrow \quad \swarrow{\scriptstyle d} & \\ & C(G) & \end{array}$$

Proposition 1.10. *The homomorphism χ is an isomorphism. The homomorphisms ϕ and d are monomorphisms. Moreover, there exists an integer q such that $q(C(G)/A(G)) = 0$.*

The above proposition makes sure that one can describe the prime ideals of $A(G)$ as pullbacks from $C(G)$.

Definition 1.11. Let p be a prime or 0. We let

$$q(H,p) := \{\alpha | \phi_H(\alpha) \equiv 0 \mod p\}$$

which is a prime ideal of $A(G)$.

The only proper inclusion of prime ideals are of the form $q(H,0) \subset q(H,p)$, which implies that $A(G)$ has Krull dimension 1.

Throughout these talks, we will focus on those ideals of the form $q(H,0)$. We list some properties of these ideals.

Proposition 1.12. *Let $q = q(H,0)$ for a subgroup H of G.*

(1) If $H \lhd J$ and J/T is a torus, then $q = q(J,0)$.
(2) Up to conjugacy, there is a uniqure subgroup $K \in \Xi(G)$ such that $q = q(K,0)$, $H \lhd K$ and K/H is a torus.
(3) For $J, H \in \Xi(G)$, $q(H,0) = q(J,0)$ if and only if $H = J \in \Xi(G)$.

Though the inner structure of $A(G)$ is complicated, its localization at prime and rationalization are easy to describe. We have:

Proposition 1.13. *Let $H \in \Xi(G)$. Then:*

(1) The localization of $A(G)$ at $q(H,0)$ is the canonical morphism

$$A(G) \to (A(G)/q(H,0))_0 \simeq \mathbb{Q}$$

(2) $\phi\colon A(G) \to C(G)$ induces an isomorphism of rationalizations.

Recall that a group is called *p-perfect* if it has no non-trivial quotient p-groups. We use H_p to denote the maximal p-perfect subgroup of H. The following result is highly extraordinary and should be used as a black box.

Theorem 1.14. *[<**empty citation**>] Let G be a finite group. Let L be a p-perfect subgroup of G that is maximal in the set of p-perfect subgroups H such that $q(H,p) = q(L,p)$ and is not a limit of such p-perfect subgroups. Then*

(1) There exists an idempotent e_L^G in $A(G)_{(p)}$, which is characterized by

$$\phi_H(e_L^G) = \begin{cases} 1 & \text{if } H_p \text{ is conjugate to } L \\ 0 & \text{otherwise} \end{cases}$$

such that $e_L^G A(G)_{(p)}$ is isomorphic to the localization of $A(G)$ at $q(L,p)$.

(2) We have

$$A(G)_{(p)} \cong \prod_L e_L^G A(G)_{(p)}$$

(3) Let E and X be G-spectra. Let $\alpha \in \mathrm{RO}(G)$, $\beta = \mathrm{Res}^G_{NL}\alpha \in \mathrm{RO}(NL)$ and $\gamma = \beta^L \in \mathrm{RO}(WL)$. Then the restriction and (geometric) fixed points functor induces isomorphisms

$$e_L^G E_\alpha^G(X) \xrightarrow{\cong} e_L^{NL} E_\beta^G(X) \xrightarrow{\cong} e_1^{WL} E_\gamma^G(X)$$

In particular, we have $e_L^G A(G)_{(p)} \cong e_L^{NL} A(NL)_{(p)} \cong e_1^{WL} A(WL)_{(p)}$.

1.2. **The Segal conjecture.** In this part, we state the Segal conjecture and sketch the proof of the conjecture in a classical fashion. Throughout this part, we will follow May's notations and generalize to the family settings.

From now on, let us fix a finite group G. We will use E_G to denote the suspension spectrum of EG_+ and $E_{\mathscr{F}}$ to denote the suspension spectrum of $E\mathscr{F}_+$ for a family $\mathscr{F}$. A family $\mathscr{F}$ determines an ideal $I\mathscr{F}$ in $A(G)$ by

$$I\mathscr{F} = \bigcap_{H\in\mathscr{F}} \ker(A(G) \to A(H)) \cong \bigcap_{H\in\mathscr{F}} q(H,0)$$

Notation 1.15. Let I be a finitely generated ideal in $\pi_0^G(\mathbb{S})$. We use $\mathbb{S}[I^{-1}]$ to denote its localization at I. We have a fiber sequence

$$\Omega\mathbb{S}/I \to \mathbb{S} \to \mathbb{S}[I^{-1}]$$

By an equivariant generalization of [Bou79, Proposition 2.5], for any G-spectrum E we have

$$E_I^\wedge \simeq \mathrm{map}(\Omega\mathbb{S}/I, E)$$

and the canonical map $E \to E_I^\wedge$ is induced by $\Omega\mathbb{S}/I \to \mathbb{S}$.

For any $\alpha\colon \mathbb{S} \to \mathbb{S} \in I\mathscr{F} \subset A(G)$, α is H-null homotopy for any $H \in \mathscr{F}$ by definition. This implies that $\mathbb{S}[I^{-1}]$ is contractible and $\Omega\mathbb{S}/I\mathscr{F}$ is H-equivalent to $\mathbb{S}$ for any $H \in \mathscr{F}$. Therefore, we have a standard map

$$\xi\colon \Sigma^\infty E\mathscr{F}_+ \to \Omega\mathbb{S}/I\mathscr{F}$$

For any G-spectrum E, ξ induces a canonical map

$$\xi^*\colon E_{I\mathscr{F}}^\wedge \simeq \mathrm{map}(\Omega\mathbb{S}/I\mathscr{F}, E) \to \mathrm{map}(E\mathscr{F}_+, E)$$

which can be view as a comparison between algebraic and geometric completion of E.

Theorem 1.16 (The Segal conjecture)**.** *Let G be a finite group and $\mathscr{F}$ be a family in G. Then*

$$\xi^*\colon \mathbb{S}_{I\mathscr{F}}^\wedge \to \mathrm{map}(E\mathscr{F}_+, \mathbb{S})$$

is an equivalence of G-spectra.

The remainder of this section will be devoted to sketch the classical methods of the proof. The proof is devided as a few steps. The first step is to note that similar to the non-equivariant case, for a bounded below and finite type spectrum E, by induction up along the Postinikov tower, one can conclude that all homotopy groups of $E_I^\wedge$ are isomorphic to the I-adic completion of the homotopy groups of E. Therefore, it suffices to prove

Theorem 1.17. *Let G be a finite group and $\mathscr{F}$ be a family in G. Then the map $E\mathscr{F} \to *$ induces an isomorphism*

$$\pi_G^*(\mathbb{S}) \xrightarrow{\cong} \pi_G^*(E\mathscr{F})$$

The Theorem 1.17 follows from a general fact. Since we are now dealing with families in G, we will first need some notations.

Definition 1.18. Let $\mathscr{F}$ be a family in G. Let $f\colon X \to Y$ be a G-map between G-spaces.

(1) We say that f is an $\mathscr{F}$-equivalence if f induces an equivalence on H-fixed points.
(2) We say that X is $\mathscr{F}$-contractible if X^H is contractible for every $H \in \mathscr{F}$.

By induction of the number of cells, one can show that $\pi_G^*(E\mathscr{F})$ is annihilated by some power of $I\mathscr{F}$. This makes sure that $\pi_G^*(E\mathscr{F})$ is itself $I\mathscr{F}$-complete. It is now converted to show the following slogan:

$$\text{An } \mathscr{F}\text{-equivalence } f\colon X \to Y \text{ induces an equivalence } \pi_G^*(f)^\wedge_{I\mathscr{F}}.$$

We will show a more general fact. Recall that we have described the prime ideals of $A(G)$, which are those $q(H,0)$ and $q(H,p)$. We define the *support* of $P \in \mathrm{Spec}(A(G))$ as the conjugacy class (L) such that P is in $\mathrm{Spec}(A(L))$ but not in $\mathrm{Spec}(A(K))$ for any subgroup K of L. Equivalently, we have

$$\mathrm{Supp}(P) = \begin{cases} (H) & \text{for } H = q(H,0) \\ (H_p) & \text{for } H = q(H,p) \end{cases}$$

Theorem 1.19. *Let S be a multiplicative set and I be an ideal in $A(G)$. Then the cohomology theory $S^{-1}\pi_G^*(-)^\wedge_I$ is $\mathscr{G}$-invariant, where*

$$\mathscr{G} = \bigcup \{\mathrm{Supp}(P) | P \cap S = \emptyset \text{ and } I \subset P\}$$

Specialization to the case where $S = \emptyset$ and $I = I\mathscr{F}$ gives us the desired result. To prove Theorem 1.19, we reduce to prove the following theorem, which was proven by Carlsson.

Theorem 1.20. *Let p be a prime and let G be a finite p-group. Then the cohomology theory $\pi_G^*(-)^\wedge_p$ is e-invariant. That is, it vanishes on non-equivariantly contractible G-spaces.*

Proof of Theorem 1.19 from Theorem 1.20. We now assume Theorem 1.20. We need to show that $S^{-1}\pi_G^*(X)^\wedge_I = 0$ if X^L is contractible for every $\mathscr{G}$-contractible space X. By general algebra fact, we need only to show that $S_P^{-1}\pi_G^*(X)^\wedge_P = 0$ if X^L is contractible for $L \in \mathrm{Supp}(P)$. By Theorem 1.14, there exists an idempotent $e_L^G \in A(G)_{(p)}$ such that $S_P^{-1}\pi_G^*(X) = e_L^G\pi_G^*(X)_{(p)} \cong e_1^{WL}\pi_{WL}^*(X^L)_{(p)}$. Let VL denote the Sylow p-subgroup of WL. Then $e_1^{WL}\pi_{WL}^*(X^L)_{(p)}$ is a direct summand of $\pi_{VL}^*(X^L)_{(p)}$. By Theorem 1.20, $\pi_{VL}^*(X^L)_{(p)}$ vanishes. Then we win. □

We now reduce to show Theorem 1.20. To prove this, we first need some notations.

Notation 1.21. We use $\tilde{E}G$ to denote the cofiber of $\Sigma_+^\infty EG \to \mathbb{S}$. It is the unbased suspension of EG. Let E, Y and X be G-spectra and $q \in \mathrm{RO}(G)$. Then we denote

$$E_G^q(X;Y) = [X, Y \wedge E]_q$$

Therefore, the cofiber sequence $\Sigma_+^\infty EG \to \mathbb{S} \to \tilde{E}G$ gives rise to a long exact sequence

$$\cdots \to E_G^q(Y;\Sigma_+^\infty EG) \to E_G^q(Y) \to E^q(Y;\tilde{E}G) \xrightarrow{\delta} E^{q+1}(Y;\Sigma_+^\infty EG) \to \cdots$$

Lemma 1.22. *Let $\mathscr{P}$ be the family of proper subgroups of G and let E be a G-spectrum. Assume that $E^*_H(-)$ is e-invariant for all $H \in \mathscr{P}$. Then $E^*_G(-)$ is e-invariant if and only if $E^*_G(\tilde{E}\mathscr{P}) = 0$.*

Proof. Let X be e-contractible and assume that $E^*_G(\tilde{E}\mathscr{P}) = 0$. We let $Y = \tilde{E}\mathscr{P}$ and Z be the (unstable) cofiber Y/S^0. We have the cofiber sequence

$$X \to X \wedge Y \to X \wedge Z$$

We need to show $E^*_G(X \wedge Y) = 0$ and $E^*_G(X \wedge Z) = 0$. We claim that $E^*_G(W_1 \wedge Y) = 0$ for every G-CW complex W_1 and $E^*_G(X \wedge W_2) = 0$ for every G-CW complex W_2 such that $W_2^G = *$. To prove this, it suffices to prove for $W_1 = G/H$ and $W_2 = G/L$ with $L \neq G$. This follows from induction and hypothesis. □

Inductively, we reduce to prove that $E^*_G(E\mathscr{P}) = 0$ for $E = \mathbb{S}_p$. In the viewpoint of Notation 1.21, we must prove that the connection map δ is an isomorphism. For a normal subgroup K of H with quotient J, we write $E^*_{H/K} = E^*_J$ for the cohomology theory represented by $\Phi^K(E_H)$.

We are going to prove the following theorems.

Theorem 1.23. *Assume that E^*_J is e-invariant for every proper subquotient J of G and let $Y = \tilde{E}\mathscr{P}$.*

(i) If G is not elementary Abelian, then $E^*_G(Y; \tilde{E}G) = 0$.
(ii) If $G = (\mathbb{Z}/p)^r$, then $E^*_G(Y; \tilde{E}G)$ is the direct sum of $p^{r(r-1)/2}$ copies of $\Sigma^{r-1}E^*_{G/G}(\mathbb{S})$.

Remark 1.24. In general, the non-equivariant cohomology theory $E^*_{G/G}$ is quite different from the underlying non-equivariant theory E^*_e.

Recall that a G-spectrum E is said to be *split* if E is a retract of E^G as naive spectra.

Theorem 1.25. *Assume that E is split, bounded below and let $Y = \tilde{E}\mathscr{P}$.*

(i) If G is not elementary Abelian, then $E^*_G(Y; EG_+) = 0$.
(ii) If $G = (\mathbb{Z}/p)^r$ and E_e has finite-dimension cohomology. Then $E^*_G(Y; EG_+)$ is the direct sum of $p^{r(r-1)/2}$ copies of $\Sigma^r E^*_e(\mathbb{S})$.

Inductively, by Theorem 1.23 and Theorem 1.25, we have:

Theorem 1.26. *Suppose that G is not elementary Abelian. Assume*

(1) E^*_J is e-invariant for all elementary Abelian subquotients J.
(2) E_J is split and $E_{K/K}$ is bounded below for all non-elementary Abelian quotient $J = H/K$.

Then E^*_J is e-invariant for all subquotients J.

It now suffices to prove that $E^*_G(E\mathscr{P}) = 0$ for $E = \mathbb{S}_p$ when $G = (\mathbb{Z}/p)^r$. By induction, suppose that the conjecture has been proven for $0 \leqslant i < r$. We will prove the connection map

$$\delta\colon E^q(Y; \tilde{E}G) \to E^{q+1}(Y; \Sigma^\infty_+ EG)$$

is an isomorphism. By Theorem 1.23 and Theorem 1.25, δ is a $\pi^*_e(\mathbb{S})$-module map between sums of same numbers of copies of $\Sigma^r\pi^*_e(\mathbb{S})$. It suffices to show that it is a bijection in degree $r-1$. Since everything locates in p-complete category, we reduce to the following lemma.

Lemma 1.27. *Let $G = (\mathbb{Z}/p)^r$ and δ as above. Then δ is a monomorphism after mod p.*

Proof. We let F be the spectrum $F = \mathrm{map}(\Sigma^\infty_+ EG, \mathrm{H}\mathbb{F}_p)$. For every (non-equivariantly) contractible space X, $F^*_G(X)$ vanishes. Let $\eta\colon \mathbb{S} \to \mathrm{H}\mathbb{F}_p$ denote the unit map. Then we have a commutative square

$$\begin{array}{ccc} \pi^{r-1}_G(Y;\tilde{E}G) & \xrightarrow{\delta} & \pi^r_G(Y;EG) \\ \downarrow{\scriptstyle \eta^*} & & \downarrow{\scriptstyle \eta^*} \\ F^{r-1}_G(Y;\tilde{E}G) & \xrightarrow[\delta]{} & F^r_G(Y;EG) \end{array}$$

where the bottom map is an isomorphism and the horizontal maps are monomorphisms. □

It now suffices to prove Theorem 1.23 and Theorem 1.25. The first theorem follows from an essentially non-stable argument.

Construction 1.28. Let G be a finite group. We let $\mathscr{S}(G)$ be the 1-category of non-trivial elementary Abelian subgroups of G with morphisms the op-inclusions. It is a G-category, where the G action is given by conjugation. For a G-space X, we let $\mathscr{S}(G)[X]$ be the category, whose objects are pairs (A, x) with $A \in \mathscr{S}(G)$ and $x \in X^A$. There is a morphism $(A, x) \to (B, y)$ if $B \subset A$ and $y = x$. The G-action on $\mathscr{S}(G)[X]$ is given by $g(A, x) = (gAg^{-1}, gx)$. Note that $\mathscr{S}(G)[*]$ is simply $\mathscr{S}(G)$.

Lemma 1.29. *The geometric realization of the category $\mathscr{S}(G)[*] = \mathscr{S}(G)$ is contractible.*

Proof. Let F_0 be the identity functor on $\mathscr{S}(G)$, $F_1\colon \mathscr{S}(G) \to \mathscr{S}(G)$ be the functor that sends A to AC, where C is a central subgroup of order p, and $F_2\colon \mathscr{S}(G) \to \mathscr{S}(G)$ be the constant functor to C. The diagram $A \leftarrow AC \to C$ gives a diagram of natural transformations $F_0 \leftarrow F_1 \to F_2$. After applying the geometric realization functor, this gives a retraction of $|\mathscr{S}(G)|$. □

Remark 1.30. Using the same strategy, we can also prove that the geometric realization of $\mathscr{S}(G)[X]$ is G-equivalent to the singular subspace of X, that is, the subspace of points with non-trivial isotropy groups.

Definition 1.31. Let $\mathscr{S}(G)[X]$ be defined as above.

(1) We define $S\colon G\mathcal{S} \to G\mathcal{S}$ by $SX := \mathscr{S}(G)[X]/\mathscr{S}(G)[*]$.
(2) Let $r = \mathrm{rank}(G)$, that is, the maximal length of strictly ascending chains $(A_0 \cdots, A_k)$ of non-trivial elementary Abelian subgroups in G. Then by construction, we have a filtration of S:

$$* = F_{-1}S \to F_0S \to F_1S \to \cdots \to F_{r-1}S = S$$

with the graded pieces

$$F_kS/F_{k-1}S(X) = \bigvee_\omega \Sigma^q(G_+ \wedge_{H(\omega)} X^{A(\omega)})$$

where ω runs over the conjugacy class of chains $(A_0, \cdots, A_k)$, $H(\omega) = \{g | gA_ig^{-1} = A_i; 0 \leqslant i \leqslant k\}$ and $A(\omega) = A_k$.

Proof of Theorem 1.23. Since SX is identified with the singular subspace of X. We have

$$E^*_G(Y;\tilde{E}G) = [\Sigma^\infty_+ X, \tilde{E}G \wedge E]_* \cong [\Sigma^\infty_+ SX, E]_*$$

while the right side is computed through the filtration F^kSX. By construction, we have

$$[\Sigma^\infty_+(F_kS/F_{k-1}S)X, E]_* \cong \bigvee_\omega [\Sigma^\infty_+ G \wedge_{H(\omega)} Y^{A(\omega)}, E]_{*+k} \cong \bigvee_\omega E^*_{H(\omega)/A(\omega)}(Y^{A(\omega)})$$

we note that when G is not elementary Abelian, or $G = (\mathbb{Z}/p)^r$ and $k \leqslant r-2$, then $H(\omega/A(\omega))$ is always a proper subquotient. By hypothesis, the right side vanishes. It remains to show the case when $G = (\mathbb{Z}/p)^r$ and $k = r-1$. In this case $A(\omega) = H(\omega) = G$ for all ω, $Y^G = S^0$ and we have $r(r-1)/2$ chains. Then the theorem follows at once. □

We now sketch the proof of Theorem 1.25. The splitting hypothesis allows us to reduce to the non-equivariant case, where we can compute the desired cohomology through Adams spectral sequence. Let V be the reduced regular G-complex representation. Then $\operatorname{colim}_n S^{nV}$ gives a model for $Y = \tilde{E}\mathscr{P}$ since it has contractible H-fixed points for all $H \in \mathscr{P}$ and $V^G = 0$.

Proof idea of Theorem 1.25. We consider V as above. Since E splits, then we have

$$E^*_G(S^V; EG_+) = [S^V, E \wedge EG_+]_* \cong [\mathbb{S}, E \wedge EG_+ \wedge S^{-V}]_* = E^G_*(EG_+ \wedge S^{-V})$$

Since E splits, we have

$$E^G_*(EG_+ \wedge S^{-V}) \cong E_*(EG_+ \wedge_G S^{-V}) \cong E_*(\mathrm{Th}(-V))$$

where $\mathrm{Th}(-)$ denotes the equivariant Thom-spectra functor. As a consequence, suppose we take $\operatorname{colim}_n S^{nV}$ as a model for Y, we have

$$E^{-*}_G(Y; EG_+) \cong \lim_n E^*_G(S^{nV}; EG_+) \cong \lim_n E_*(\mathrm{Th}(-nV)) \cong \lim_n \pi_*(X_n)$$

where $X_n = E \wedge \mathrm{Th}(-nV)$. The connection map is given in a canonical way.

Now we consider the Adams spectral sequence that computes $\pi_*(X_n)$. Let $\mathrm{H}^*(-)$ denote the mod p cohomology. We have a sequence of Adams spectral sequences

$$E_2 = \mathrm{Ext}_{\mathcal{A}_p}(\mathrm{H}^*(X_n), \mathbb{F}_p)$$

with the limit

$$E_2 = \lim_n \mathrm{Ext}_{\mathcal{A}_p}(\mathrm{H}^*(E) \otimes \mathrm{H}^*(\mathrm{Th}(-\mathrm{nV})), \mathbb{F}_p)$$

that calculates $E^{-*}_G(Y; EG_+)$. The connection map in this sequential limit is given by

$$\chi(V)\colon \mathrm{H}^*(\mathrm{Th}(-\mathrm{nV})) \to \mathrm{H}^*(\mathrm{Th}(-(\mathrm{n}+1)\mathrm{V}))$$

By some geometrical reasons, $\chi(V)$ is nilpotent if G is not elementary Abelian. This implies that $E^{-*}_G(Y; EG_+) = 0$ when G is not elementary Abelian.

Now suppose that $G = (\mathbb{Z}/p)^r$. Then $\chi(V)$ determines an element $v \in \mathrm{H}^*(BG)$, so we have

$$\operatorname*{colim}_n \mathrm{H}^*(\mathrm{Th}(-\mathrm{nV})) \cong \mathrm{H}^*(BG)[v^{-1}]$$

It then can be calculated that for any finite dimensional $\mathcal{A}_p$-module K, the second page

$$\mathrm{Ext}_{\mathcal{A}_p}(K \otimes \mathrm{H}^*(BG)[v^{-1}], \mathbb{F}_p)$$

concentrates in homological degree $-r$ and has dimension $p^{r(r-1)}/2$. By induction, let W denote the wedge of $p^{r(r-1)}/2$-copies of S^{-r}, the map $W \to \mathrm{Th}^{-nV}$ induces an isomorphism on cohomology. □

1.3. **Tate construction.** Now we explain a classical corollary of the Segal conjecture, which states that the C_p-Tate spectrum of $\mathbb{S}$ is identified with $\mathbb{S}_p$. This plays a critical role in the formation of topological cyclic homology in [NS18]. Our Tate construction is much more classical and mostly in the (genuine) equivariant world. As expected, it is proved to coincide with that of [Lur17] and [NS18] in a strong sense.

Let G be a finite group, in which case the Segal conjecture is valid, though a compact Lie group is enough for forming the definition. For any G-spectrum E, the projection $EG_+ \to S^0$ induces a canonical map

$$\varepsilon\colon E \simeq \mathrm{map}(\mathbb{S}, E) \to \mathrm{map}(EG_+, E)$$

After smashing with the fiber sequence $EG_+ \to \mathbb{S} \to \tilde{E}F$, we obtain a commutative diagram:

$$\begin{array}{ccccc} E\wedge EG_+ & \longrightarrow & E & \longrightarrow & E\wedge \tilde{E}G \\ \downarrow{\scriptstyle \varepsilon\wedge\mathrm{id}} & & \downarrow{\scriptstyle \varepsilon} & & \downarrow{\scriptstyle \varepsilon\wedge\mathrm{id}} \\ \mathrm{map}(EG_+,E)\wedge EG_+ & \longrightarrow & \mathrm{map}(EG_+,E) & \longrightarrow & \mathrm{map}(EG_+,E)\wedge \tilde{E}G \end{array}$$

It can be checked directly that the left vertical arrow is always an equivalence.

Definition 1.32. Let G be a finite group and E be a G-spectrum. We define $T_G(E) := \mathrm{map}(EG_+, E)\wedge \tilde{E}G$ and call it the *G-equivariant Tate spectrum* of E.

Remark 1.33. The above definition is still valid for any compact Lie group G.

We can also apply the G-homotopy fixed points functor to the above diagram. After applying $(-)^{hG}$ to the underlying naive spectrum, all vertical arrows become equivalences and the fiber sequence becomes

$$E_{hG} \to E^{hG} \to E^{tG}$$

We call E^{tG} the *Tate spectrum* of E. This coincide with the definition of [NS18].

Corollary 1.34. *Let G be a finite group and $\mathbb{S}$ be the G-sphere, then we have an equivalence*

$$T_G(\mathbb{S}) \simeq (\Sigma^\infty \tilde{E}G)^\wedge_I$$

In particular, if we take $G = C_p$ and apply the homotopy fixed points functor to the underlying naive spectrum, we obtain an equivalence

$$\mathbb{S}^{tC_p} \simeq \mathbb{S}_p$$

Proof. By the Segal conjecture, we have

$$T_G(\mathbb{S}) \simeq \mathrm{map}(EG_+, \mathbb{S})\wedge \tilde{E}G \simeq \mathbb{S}^\wedge_I \wedge \tilde{E}G \simeq (\Sigma^\infty \tilde{E}G)^\wedge_I \qquad \square$$

2. Recollection of synthetic homotopy theory

In this part, we recall several facts we may use in the synthetic homotopy theory for the proof of generalized Segal conjecture.

Definition 2.1 ([Pst23])**.** Let E be an Adams type ring spectrum. The category of synthetic spectrum based on E is defined as

$$\mathrm{Syn}_E := \mathrm{Shv}_\Sigma(\mathrm{Sp}^{\mathrm{fp}}_E, \mathrm{Sp})$$

Here $\mathrm{Sp}_E^{\mathrm{fp}}$ is the category of finite E-projective spectra and is equipped with the Grothendieck topology generated by covering families consisting of a single E_*-surjective map.

Intuitively, we can view Syn_E as the "derived category with respect to E_*-homology". There is an adjunction

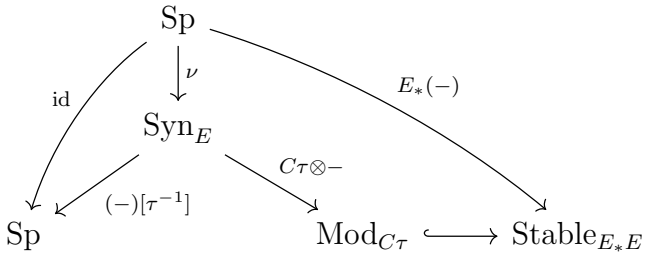
$$\Sigma_+^\infty \colon \mathrm{Shv}_\Sigma(\mathrm{Sp}_E^{\mathrm{fp}}) \rightleftharpoons \mathrm{Syn}_E : \Omega^\infty$$

This results in a synthetic analogue functor $\nu \colon \mathrm{Sp} \to \mathrm{Syn}_E$, which is obtained by taking X to the sheafification of $\tau_{\geq 0}\mathrm{map}(-, X)$. A formal argument shows that Syn_E is equipped with a Day convolution symmetric monoidal structure and ν is lax monoidal.

Definition 2.2. We define the bigraded sphere $\mathbb{S}^{a,b} := \Sigma^{-b}\mathbb{S}^{a+b}$. There is a map $\tau \colon \mathbb{S}^{0,-1} \to \mathbb{S}^{0,0}$ defined as the comparison:

$$\mathbb{S}^{0,-1} = \Sigma\nu\mathbb{S}^{-1} \to \nu\Sigma\mathbb{S}^{-1} = \mathbb{S}^{0,0}$$

Using τ and both its generic and special fibers, we can summarize everything in the following picture:



We list some properties of synthetic spectra.

Proposition 2.3. *Let E be an Adams type ring spectrum and let Syn_E be defined as above.*

1. *The synthetic analogue functor ν is fully faithful and preserves filtered colimits.*
2. *Restricting to $\mathrm{Sp}_E^{\mathrm{fp}}$, the synthetic analogue functor ν is strong symmetric monoidal.*
3. *For a cofiber sequence $X \to Y \to Z$ in Sp, $\nu X \to \nu Y \to \nu Z$ is a cofiber sequence if and only if*
$$0 \to E_*X \to E_*Y \to E_*Z \to 0$$
is a short exact sequence.
4. *The cofiber $C\tau$ admits an $\mathbb{E}_\infty$ structure.*
5. *Suppose that E is a homotopy commutative ring spectrum, the inclusion $\mathrm{Mod}_{C\tau} \to \mathrm{Stable}_{E_*E}$ is fully faithful. As a consequence, we have*
$$\pi_{t-s,s}(C\tau \otimes \nu X) \cong \mathrm{Ext}_{E_*E}^{s,t}(E_*, E_*X)$$
6. *For any $Y \in \mathrm{Sp}$, Y is E-nilpotent complete if and only if νY is τ-complete. That is, we have an equivalence*
$$\nu Y \to \lim_n \nu Y \otimes C\tau^n$$

We will use the following technical lemma in order to prove the Segal conjecture for large objects.

Lemma 2.4. *Let $F \colon J \to \mathrm{Sp}$ be a diagram of spectra. Then the coassembly map*

$$\Big(\lim_J \nu F(j)\Big)[\tau^{-1}] \to \lim_J \Big(\nu(F(j))[\tau^{-1}]\Big) \simeq \lim_J F(j)$$

is an equivalence.

Proof. Using the compactness of bigraded sphere and fully faithfulness of ν, we have

$$\begin{aligned}\mathrm{Map}_{\mathrm{Sp}}(\mathbb{S}^n, \Big(\lim_J \nu F(j)\Big)[\tau^{-1}]) &\simeq \mathrm{colim}_t \mathrm{Map}_{\mathrm{Syn}_E}(\Sigma^{0,-t}\nu\mathbb{S}^n, \lim_J \nu F(j)) \\ &\simeq \mathrm{colim}_t \lim_J \mathrm{Map}_{\mathrm{Syn}_E}(\Sigma^{0,-t}\nu\mathbb{S}^n, \nu F(j)) \\ &\simeq \mathrm{colim}_t \lim_J \Omega^t \mathrm{Map}_{\mathrm{Syn}_E}(\nu\mathbb{S}^{n-t}, \nu F(j)) \\ &\simeq \mathrm{colim}_t \lim_J \Omega^t \mathrm{Map}_{\mathrm{Sp}}(\mathbb{S}^{n-t}, F(j)) \\ &\simeq \mathrm{colim}_t \lim_J \mathrm{Map}_{\mathrm{Sp}}(\mathbb{S}^{n}, F(j)) \\ &\simeq \mathrm{Map}_{\mathrm{Sp}}(\mathbb{S}^n, \lim_J F(j)) \end{aligned}$$

□

3. Segal conjecture for large objects

In this section, we explain the proof of the following Segal conjecture for larger objects:

Theorem 3.1. *Let I be the augmented ideal of the Steenrod algebra $\mathcal{A}_p$. Let X be a p-complete bounded below spectrum. If X is I-complete, then the canonical map*

$$X \to X^{tC_p}$$

is an equivalence.

In order to prove this theorem for a more general X, we need to trace the outline of the classical proof in the first section. Recall that in Section 1, the proof is outlined by analyzing the G-equivariant E-cohomology for $Y = \tilde{E}\mathscr{P}$, where $\mathscr{P}$ is the proper family of G. In order to generalize this theorem, we need to replace the G-sphere $\mathbb{S}_G$ by X, the spectrum Y then automatically changes. When we only care about the case $G = C_p$, we have $E\mathscr{P} = EC_p$ and we need to replace Y by $Y' = \tilde{E}C_p \wedge X$. The upshot is Theorem 1.23 and Theorem 1.25 since the other arguments are all formal. Theorem 1.23 works well for this new Y' because it follows from an unstable argument for general spaces. Therefore, the critical argument concentrates in Theorem 1.25.

The proof of Theorem 1.25 follows from an inspection of Adams spectral sequence. To generalize this spectral sequence argument for X, we need to replace the cohomology of sphere $\mathbb{F}_p$ by $\mathrm{H}^*(X)$. Of course one can move $\mathrm{Th}(-nV)$ to the second term. Then the argument reduces to the following:

Theorem 3.2. *Let V be the regular complex representation of G and let X satisfies the condition in Theorem 3.1. The natural map*

$$\mathrm{H}_*(X) \to \lim_k \mathrm{H}_*(X) \otimes \mathrm{H}_*(\Sigma\mathrm{Th}(kV))$$

is an isomorphism.

Proof of Theorem 3.1. Consider the following cofiber sequence:

$$\lim_k (\mathbb{S}^{kV} \otimes X)_{hC_p} \to \lim_k (\mathbb{S}^{kV} \otimes X)^{hC_p} \to \lim_k (\mathbb{S}^{kV} \otimes X)^{tC_p}$$

The first term is identified with $\lim_k(\Sigma^\infty \mathrm{Th}(kV) \otimes X)$. Moreover, since the underlying non-equivariant connection map in the middle term is null, then we conclude that the middle

term vanish. The cofiber of the connection maps are simply $\vee_{1,2}\Sigma^\infty_+ C_p \wedge X$, which is free so admits trivial tate construction. As a consequence, we have

$$\lim_k X \otimes \Sigma^\infty \Sigma \mathrm{Th}(kV) \simeq \lim_k (\mathbb{S}^{kV} \otimes X)^{tC_p} \simeq X^{tC_p}$$

which by Lemma 2.4 fits into the following commutative diagram:

$$\begin{array}{ccc} (\nu X)[\tau^{-1}] & \longrightarrow & \left(\lim_k \nu(X \otimes \Sigma \mathrm{Th}(kV))\right)[\tau^{-1}] \\ \cong\downarrow & & \downarrow\cong \\ X & \longrightarrow & X^{tC_p} \end{array}$$

Suppose that we have already proved Theorem 3.2. Then since we have

$$C\tau \otimes \nu X \simeq \mathrm{H}_*(X) \simeq \lim_k \mathrm{H}_*(X) \otimes \mathrm{H}_*(\Sigma \mathrm{Th}(kV)) \simeq C\tau \otimes \left(\lim_k \nu(X \otimes \Sigma \mathrm{Th}(kV))\right)$$

Note that X is p-complete and bounded below implies both νX and $\lim_k \nu(X \otimes \mathrm{Th}(kV))$ are τ-complete. Therefore, we have $\nu X \simeq \lim_k \nu(X \otimes \mathrm{Th}(kV))$, which implies the upper arrow in the diagram to be an equivalence, proving the theorem. □

The proof of Theorem 3.2 will demand the following lemma, which is a very standard homological algebra exercise.

Lemma 3.3. *Let $X \in \mathrm{Stable}(\mathcal{A}_p)^\heartsuit$ be a finite type $\mathcal{A}_p$-comodule. Let $F\colon J \to \mathrm{Stable}(\mathcal{A}_p)^\heartsuit$ be a diagram of $\mathcal{A}$-comodules where the internal degree of $F(j)$ are uniformly bounded below. Then the natural comparison map*

$$X \otimes \left(\lim_J F(j)\right) \to \lim_J X \otimes F(j)$$

is an equivalence.

Proof of Theorem 3.2. By hypothesis, since $\mathrm{H}_*(X)$ is I-complete, we need only to show the isomorphism on the associated graded pieces. That means, we take $K_n = \mathrm{fib}(\mathrm{H}_*(X)/I^{n+1} \to \mathrm{H}_*(X)/I^n)$. We need to show that we have a natural isomorphism

$$K_n \to \lim_k K_n \otimes \mathrm{H}_*(\Sigma \mathrm{Th}(kV))$$

Since $\mathcal{A}_p$ acts trivially on K_n, then we have $K_n \cong \bigoplus_i \mathbb{F}_p[i]^{\oplus J_i}$, which is a retract of $\prod_i \mathbb{F}_p[i]^{\times J_i}$. It now suffices to show that we have

$$\prod_i \mathbb{F}_p[i]^{\times J_i} \to \lim_k \left((\prod_i \mathbb{F}_p[i]^{\times J_i}) \otimes \mathrm{H}_*(\Sigma \mathrm{Th}(kV))\right)$$

to be an isomorphism. By Lemma 3.3, it suffices to show termwisely we have an isomorphism

$$\mathbb{F}_p[i]^{\times J_i} \to \lim_k (\mathbb{F}_p[i] \otimes \mathrm{H}_*(\Sigma \mathrm{Th}(kV))$$

This follows from the proof of Theorem 1.25. □

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